

DeepSeek-OCR and Document Compression

Deep Dive into DeepSeek — Chapter 18

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One page crosses **five different counts**.

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Goal: follow one inventory page from pixels to a scored typed output — encoder, view policy, feature budget, output grammar, runtime record — and keep every count and every claim attached to the layer that produced it.

The document object comes before the model

Compress before global attention

Views, budgets, and the count ledger

Training roles and reported evidence

Typed outputs, runtime paths, and records

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One page, four requested objects

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The chapter's running page I_{inv} is one schematic inventory page: a heading, a three-column table, a footer, and a declared 1000×1000 top-left-origin frame. Identical bytes, four contracts.

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The chapter's running page I_{inv} is one schematic inventory page: a heading, a three-column table, a footer, and a declared 1000×1000 top-left-origin frame. Identical bytes, four contracts.

Page identity is only the first field of an OCR result. The processor decides what visual evidence arrives; the prompt decides which artifact to emit; the scorer decides which differences matter.

The execution record, before anything is measured

Definition (OCR execution record)

$$R_{\text{ocr}} = (l, \rho, \tau, p, \mu, d, y, S)$$

l page artifact and dimensions; ρ executed resize, padding, crop and tile policy; τ the declared visual-position counter; p prompt and requested output type; μ model and revision; d runtime and decoding settings; y raw typed output; S parser and scorer.

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Changing any field that changes interpretation creates a different evaluated system.

Compress before global attention

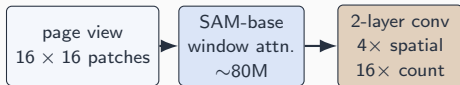
DeepEncoder orders the expensive operations

page view
 16×16 patches

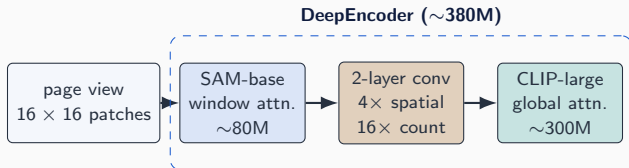
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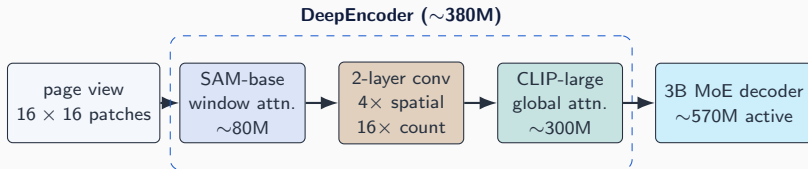
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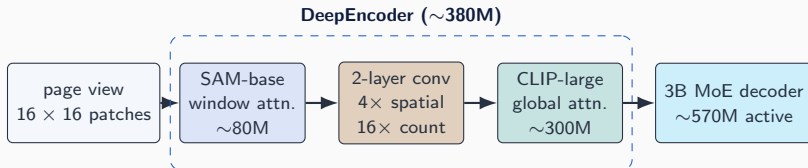
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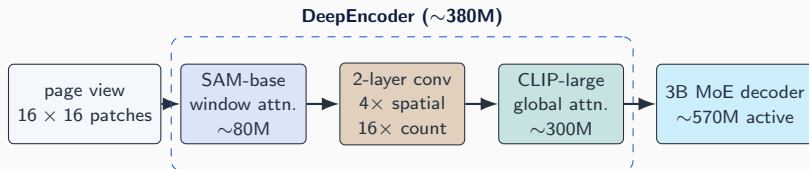


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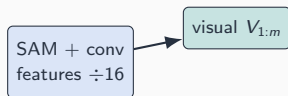
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The ordering is the design. Moving global attention before the compressor would expose it to all 4096 patch positions and defeat the activation-saving purpose. [REPORTED: DeepSeek-OCR technical report]

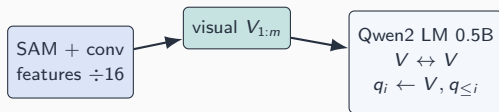
OCR-2: compression solved length, not reading order

SAM + conv
features $\div 16$

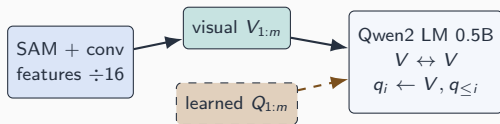
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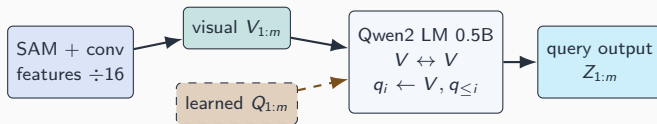
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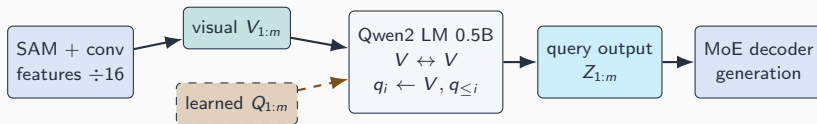
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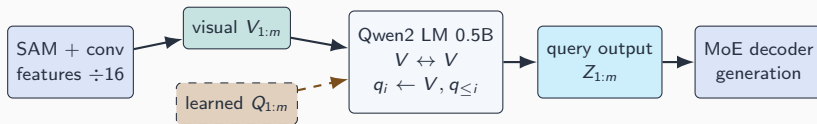
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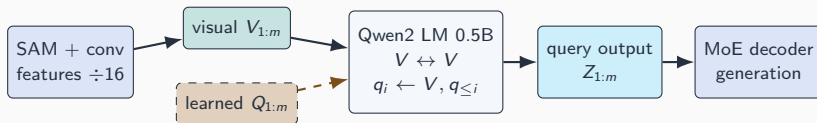


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A raster scan flattens a page top-left to bottom-right even when the semantic order moves between columns, captions, formulas, and tables. OCR-2 keeps the visual tokenizer and replaces CLIP with a Qwen2-0.5B-base encoder plus $N_q = N_v = m$ learned *causal flow queries*.

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Visual states stay bidirectional; the learned reading flow is causal; only the m query states reach the decoder. Two cascaded causal stages — order first, then generation. [REPORTED:

DeepSeek-OCR-2 technical report]

Views, budgets, and the count ledger

Native modes and dynamic budgets

For a square native side s :

$$N_{\text{native}}(s) = \frac{(s/16)^2}{16} = \left(\frac{s}{64}\right)^2 \implies 512:64, 640:100, 1024:256, 1280:400.$$

Release and mode	One local view	Global view	Four-local total
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$N_{\text{OCR}}(k) = 100k + 256$ and $N_{\text{OCR2}}(k) = 144k + 256$ with $0 \leq k \leq 6$, so OCR-2's decoder-facing budget spans 256–1120. The same local-view count does *not* mean the same budget. [REPORTED: DeepSeek-OCR and OCR-2 technical reports]

The accounting rule that keeps the counts apart

Contract (Processor accounting)

A visual-token claim must name the executed resolution mode, local-view count and grid, global view, and counting layer. Paper-level compressed features, runtime image-marker positions, separator positions, and total prompt positions must not be reported as one counter.

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A defensible row: “OCR-2 dynamic; four 768×768 local views in a 2×2 grid; one 1024×1024 global view; 832 paper-level features; runtime replacement length recorded separately.” Never “the page is 832 tokens.” [DERIVED: book accounting from reported view geometry]

Training roles and reported evidence

Two stages, then three

Release	Stage	Updated / frozen
OCR	DeepEncoder pretraining	DeepEncoder with a compact LM, next-token objective
OCR	Full-model training	freeze SAM + compressor; CLIP and MoE decoder trainable
OCR-2	Encoder pretraining	visual tokenizer + Qwen2 encoder, temporary lightweight decoder
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Because the data recipe also changed, the OCR-to-OCR-2 comparison is close to, but not literally, an [architecture-only ablation](#). [REPORTED: DeepSeek-OCR and OCR-2 technical reports]

The reported comparison, kept at its own claim level

Paper row	Max feat. ↓	Overall ↑	Text edit ↓	Read-order edit ↓
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These rows compare complete trained systems. The book did not reproduce the training run or execute OmniDocBench v1.5, so they support no local benchmark and no causal attribution to Visual Causal Flow alone.

Typed outputs, runtime paths, and records

Typed output: parser gate first, scorer second

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Recording an out-of-memory error as an empty transcription, or coercing malformed table markup into an empty grid, turns systems failures into model errors and corrupts the denominator.

From concept to code: where the extra position comes from

The paper-level equation and the runtime replacement length are computed by different code, and the runtime one adds a boundary position.

```
from DeepSeek-OCR-2/DeepSeek-OCR2-master/DeepSeek-OCR2-vllm/deepseek_ocr2.py

h = w = math.ceil((base_size // patch_size) / downsample_ratio)
h2 = w2 = math.ceil((image_size // patch_size) / downsample_ratio)
global_views_tokens = h * (w)
    local_views_tokens = (num_height_tiles * h2) * (num_width_tiles * w2)
return global_views_tokens + local_views_tokens + 1
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With `base_size = 1024`, `patch_size = 16` and `downsample_ratio = 4`: $h = w = 16$, so the global view contributes 256; `image_size = 768` gives $h_2 = w_2 = 12$, so each local view contributes 144. The trailing `+ 1` is the boundary position — that is the whole gap between 832 and 833. [INSPECTED: DeepSeek-OCR-2 vLLM integration at 2f3699ebbb96]

Records, not screenshots

Failure boundary	Retained evidence	Claim that remains valid
Load or dependency	model id, revision, environment, exception	blocked; no quality result
Processor	page hash, dimensions, config, transform error	preprocessing blocker; no token or accuracy row
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This project has not executed an OCR or OCR-2 accuracy benchmark. A score would still need the dataset manifest, references, evaluator, aggregation command, and invalid-run policy. [\[OPEN:](#)

no local OCR accuracy observation]

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- **row-separator positions in the older path** → `DeepSeek-OCR-vllm/deepseek_ocr.py`

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- **row-separator positions in the older path** → `DeepSeek-OCR-vllm/deepseek_ocr.py`

Compress the page, order the evidence, type the output, name the counter.

Next: the Systems Foundations chapter in Volume II carries these runtime implications — registration, scheduling, and measurement provenance — past the single-page record.